

Technoeconomic Assessment and Robust Validation of Hydropower Expansion Pathways under Deep Uncertainty: A Case Study of the Expansion of Bumbuna Hydroelectric Power in Sierra Leone

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Abstract

Energy planning under uncertainty remains a critical challenge in developing economies, where data limitations and climate variability complicate long, term investment decisions. In Sierra Leone, these challenges are particularly acute given low electricity access 36% and heavy reliance on hydropower with strong seasonal variability. This study addresses the limitations of deterministic energy modelling, which typically assumes perfect foresight and produces single optimal pathways that may not remain valid under uncertain futures. To overcome this, we integrate an OSeMOSYS, based technoeconomic optimization model with a Robust Decision Making (RDM) framework to evaluate resilient energy strategies. The approach is applied to the case of the Bumbuna Hydroelectric Expansion, including the proposed upstream Yiben reservoir. Key contributions include the systematic validation of least, cost model outputs under deep uncertainty through hundreds of parameter variations, and the identification of critical uncertainty drivers using scenario discovery techniques (PRIM). Results show that scenarios incorporating the Yiben reservoir (Scenarios 6 and 7) achieve up to ,100% renewable electricity generation by 2030, reduce reliance on heavy fuel oil generation, and deliver lower system costs and Levelised Cost of Electricity compared to alternatives without storage expansion. Furthermore, these scenarios consistently demonstrate robustness across a wide range of futures, clustering in low, cost, high, renewable outcome spaces. The findings highlight hydropower storage as a key enabler of system reliability and decarbonization. From a policy perspective, the study underscores the importance of integrating uncertainty analysis into national energy planning and supports prioritising cascade hydropower development to enhance energy security and resilience in Sierra Leone.

Keywords

Energy systems modelling; Robust Decision Making; OSeMOSYS; Hydropower expansion; Technoeconomic analysis; Deep uncertainty; Renewable energy planning; Sierra Leone

1. Introduction

The global transition toward sustainable energy systems is increasingly characterized by complex decision, making environments shaped by technological, economic, and policy uncertainties. Energy system optimization models have become essential tools for informing long, term planning and investment decisions; however, their outputs are often contingent on deterministic assumptions that may not adequately capture real, world variability (Weijermars *et al.*, 2012). Conventional modelling approaches tend to rely on a limited number of scenarios or predefined probability distributions, which can lead to strategies that are optimal under assumed conditions but fragile when exposed to unforeseen changes (Victor-Gallardo *et al.*, 2026). This limitation is especially critical in the context of developing countries, where energy systems are rapidly evolving and subject to deep uncertainty.

Decision Making under Deep Uncertainty (DMDU) frameworks, such as Robust Decision Making (RDM), have emerged as powerful approaches to address these challenges by shifting the focus from optimality under a single future to robustness across a wide range of plausible futures. RDM enables the systematic exploration of uncertainty spaces through computational experiments, stress, testing candidate strategies, and identifying vulnerabilities and trade, offs (Victor-Gallardo *et al.*, 2026). Recent methodological advancements, including the integration of RDM with open, source energy system optimization models such as OSeMOSYS, have further enhanced the transparency, reproducibility, and analytical depth of energy planning studies (Victor-Gallardo *et al.*, 2026). Despite these developments, a practical gap persists between advanced uncertainty analysis methods and their routine application in energy modelling workflows.

This gap is particularly evident in the context of Sub, Saharan Africa, where data limitations and rapidly changing energy demand patterns complicate long, term planning efforts. Sierra Leone, for instance, relies heavily on hydropower as a cornerstone of its electricity generation mix, with existing infrastructure such as the Bumbuna Hydroelectric Power Plant and proposed expansion projects playing a central role in national energy strategies(Kiley *et al.*, 2024). Energy modelling efforts in the country, often implemented using tools such as OSeMOSYS,

have been instrumental in assessing future capacity expansion pathways, system costs, and emissions trajectories under different scenarios (Kiley *et al.*, 2024). However, these studies frequently depend on deterministic assumptions or limited sensitivity analyses, which may not fully capture the uncertainty surrounding key parameters such as hydrological variability, demand growth, and technology costs.

Furthermore, hydropower expansion projects, including the Bumbuna II development, are associated with significant technoeconomic and environmental considerations, requiring robust evaluation frameworks that account for both system performance and uncertainty. Traditional approaches to renewable energy planning have highlighted the difficulty of balancing multiple, often conflicting criteria, such as cost, reliability, and environmental impact, under uncertain conditions (Mohammad Husain *et al.*, 2024). This underscores the need for integrated methodologies that combine energy system modelling with advanced decision, support tools capable of addressing deep uncertainty.

In response to these challenges, this study applies a Robust Decision, Making workflow to validate the outputs of energy system modelling for the Bumbuna Hydroelectric Power Expansion in Sierra Leone. By coupling technoeconomic analysis with an exploratory modelling framework, the study seeks to enhance the credibility and policy relevance of model, based insights. Specifically, it aims to assess the robustness of model outcomes across a wide range of uncertain futures, thereby providing more resilient guidance for infrastructure planning and investment decisions. In doing so, the paper contributes to the growing body of literature on uncertainty, aware energy planning and demonstrates the applicability of RDM, based approaches in data, constrained, high, impact contexts.

The paper is organised as follows. Section 1 presents the introduction and literature review. Section 2 describes the literature review methodology, including the reference energy system, modelling techniques, data description, and scenario construction and assumptions. Section 3 presents the results of the scenario analysis. Section 4 presents the discussion and policy and Section 5 concludes with key insights from the analysis.

2.1 Technoeconomic Analysis of Hydropower Systems

Hydropower serves as a critical component of renewable energy expansion in developing regions, notably Sub-Saharan Africa, due to its ability to provide reliable, low-cost electricity necessary to mitigate severe rural electrification deficits (Olkkonen *et al.*, 2023; Milambo *et*

et al., 2025). However, while untapped potential is vast, widespread deployment is frequently constrained by topography, hydrological variability, and limited financial resources (Milambo *et al.*, 2025). Furthermore, existing hydropower infrastructure in these developing areas is increasingly being evaluated alongside or hybridized with other renewable technologies to maximize resource complementarity, adapt to climate uncertainty, and reduce seasonal generation variability (Alhassan *et al.*, 2023) (Faruqui, Jawad and Masood, 2023).

To justify these massive infrastructural investments and navigate deep uncertainties, robust technoeconomic analyses are essential. A project's economic viability relies heavily on evaluating three core cost structures: capital expenditures (CAPEX), operational and maintenance expenses (OPEX), and the Levelized Cost of Electricity (LCOE) (Carapellucci, Giordano and Pierguidi, 2015) (Mulenga *et al.*, 2023). CAPEX covers substantial upfront equipment and civil works investments, whereas OPEX accounts for ongoing system maintenance and repairs (Carapellucci, Giordano and Pierguidi, 2015);(Milambo *et al.*, 2025). Despite high initial CAPEX requirements, hydropower and modern hybridized systems can achieve highly competitive LCOE rates, ensuring rapid capital recovery, system reliability, and long-term financial resilience under uncertain market conditions (Milambo *et al.*, 2025).

2.2 Energy System Optimization Models (ESOMs)

Energy System Optimization Models (ESOMs) are fundamental decision-support frameworks utilized to navigate the complexities of capacity expansion and infrastructure planning (Wali *et al.*, 2023). These models are designed to identify the most cost-effective and reliable energy transition pathways by evaluating trade-offs between future demand, technology costs, and environmental impacts. In developing regions grappling with supply deficits, ESOMs play a critical role in informing sustainable and robust long-term energy policies (Ayuketah *et al.*, 2022)

Among the various ESOMs available, the Open-Source energy MOdelling SYStem (OSeMOSYS) has emerged as a highly adaptable and widely adopted tool. OSeMOSYS is a deterministic, bottom-up linear optimization program that estimates the lowest net present value (NPV) costs required to meet exogenously defined energy demands (Gebremeskel, Ahlgren and Beyene, 2023). It evaluates complex energy balances by meticulously accounting for the costs induced by different technologies, alongside emissions and resource extraction limits (Manirambona, Talai and Kimutai, 2022). Because it is open-source and relatively

accessible, OSeMOSYS is highly advantageous for resource-constrained contexts, enabling planners to effectively incorporate unique regional characteristics such as the urban-rural divide into their models (Gebremeskel, Ahlgren and Beyene, 2023).

In Sub-Saharan Africa, OSeMOSYS has been extensively deployed to address critical electrification and capacity expansion challenges. For instance, the tool has been used to investigate mid- to long-term capacity planning and the integration of variable renewable energy in Kenya (Kihara *et al.*, 2024), as well as to evaluate decentralized electrification pathways in Tanzania (Manirambona, Talai and Kimutai, 2022). Furthermore, OSeMOSYS has been applied to optimize the long-term electricity supply mix in Ethiopia, aiding in the assessment of hydropower and other renewable resources (Gebremeskel, Ahlgren and Beyene, 2023). Despite its widespread application, these regional studies frequently rely on deterministic assumptions. This highlights a critical need to couple models like OSeMOSYS with advanced exploratory frameworks, such as Robust Decision Making (RDM), to adequately address the deep uncertainties surrounding hydrological variability and shifting technology parameters.

2.3 Decision-Making under Deep Uncertainty (DMDU)

Long-term energy infrastructure planning, particularly for massive hydropower expansions, involves highly complex socio-economic, environmental, and technological variables that are inherently unpredictable. Traditional deterministic models frequently struggle to account for these complexities and rapid shifts in demand. Consequently, Decision-Making under Deep Uncertainty (DMDU) frameworks have emerged as critical tools for strategic energy planning. These approaches recognize that the future is characterized by deep complexity and cannot be accurately predicted by a single forecasted outcome, thereby necessitating analytical methods that evaluate system vulnerabilities across a broad spectrum of plausible futures (Faryadras *et al.*, 2025). Furthermore, formally incorporating deep uncertainty into planning models is essential to help decision-makers mitigate severe risks associated with climate change variability and the long-term availability of hydrological resources (Carvajal *et al.*, 2019).

A prominent and highly effective methodology within the DMDU paradigm is Robust Decision Making (RDM). Instead of seeking a single optimal solution tailored to one "most-likely" future, RDM explicitly identifies and develops strategies that perform reasonably well across a multitude of highly uncertain scenarios (Faryadras *et al.*, 2025). In power systems heavily

reliant on variable renewable resources like hydropower, applying robust optimization and scenario-based robust planning is critical for ensuring system resilience against climatic droughts and operational bottlenecks (Roldan-Fernandez *et al.*, 2024). Ultimately, planners rely on RDM frameworks to hedge against the immense financial and structural risks of long-term infrastructure investments, purposefully seeking technology configurations that minimize power deficits even under severe, worst-case external conditions (Gonzalez-Salazar and Roger Poganietz, 2022).

The RDM workflow relies heavily on exploratory modelling to systematically define and map out deep uncertainties. Exploratory scenario planning facilitates the explicit computational analysis of uncertainties by consistently posing the exploratory question, "What might the future hold?", rather than aiming to hit a predefined normative target (Faryadras *et al.*, 2025). By coupling technoeconomic energy models with exploratory scenario generation, decision-makers can rigorously uncover the real-world implications of alternative development pathways, comprehensively assessing how critical factors like hydrological runoff and shifting technology costs will impact the grid over multiple decades (Gebremeskel, Ahlgren and Beyene, 2023). This deep integration of exploratory models provides the essential foundation needed to validate large-scale capacity expansions in high-impact, data-constrained

2.4 Scenario Discovery Methods

Addressing deep uncertainty in long-term infrastructure planning requires methodologies that can systematically explore vast decision spaces. Scenario discovery has emerged as a vital analytical approach to characterize and communicate deep uncertainty by evaluating large ensembles of simulation model outputs to identify combinations of input parameters that are strongly predictive of specified policy-relevant results (Lempert, Bryant and Bankes, no date). Rather than attempting to find a single optimal forecast, scenario discovery reveals the uncertain conditions under which a proposed policy fails or succeeds, making it a cornerstone of Robust Decision Making (RDM) frameworks (Kwakkel and Jaxa-Rozen, 2016).

A primary algorithmic tool utilized for scenario discovery is the Patient Rule Induction Method (PRIM) (Kaveh, Bakhshpoori and Hamze-Ziabari, 2016). Originally introduced by (Friedman and Fisher, no date) as a "bump-hunting" or subgroup discovery technique, PRIM searches for sub-regions—described as hyper-rectangular boxes—with a high concentration of target outcomes by iteratively peeling away small slices of data from the input space (Lempert,

Bryant and Bankes, no date). In contrast to greedier classification algorithms like CART, PRIM's "patient" optimization procedure prevents it from prematurely fragmenting data, allowing it to better balance outcome density, coverage, and rule interpretability (*Engelberg, 2024; Lempert, Bryant, & Bankes, 2008*). - *Google Scholar*, no date)

Within the domain of energy planning, scenario discovery and rule induction have proven highly effective for navigating complex, conflicting criteria under uncertainty. For instance, (Bryant and Lempert, 2010) successfully demonstrated scenario discovery by evaluating the conditions under which a 25% renewable energy policy in the United States might incur unacceptable economic costs, while (Hamarat *et al.*, 2014) used the approach to design adaptive policies steering the European energy system toward sustainability (Kwakkel and Jaxa-Rozen, 2016). Additionally, rule induction techniques have been applied to broader energy efficiency challenges to discover predictive knowledge regarding electricity customer consumption behaviours and to design targeted smart grid tariffs (Chen *et al.*, 2016). These applications underscore the capacity of scenario discovery and PRIM to provide resilient, uncertainty-aware guidance for complex energy system expansions.

2.5 Research Gap Summary

Despite the extensive application of Energy System Optimization Models (ESOMs) in developing regions, a critical research gap persists regarding the treatment of deep uncertainty in long-term infrastructure planning (Gebremeskel, Ahlgren and Beyene, 2023). Traditional energy planning frameworks frequently rely on deterministic or probabilistic assumptions, which often generate fragile strategies that fail to capture the complex, unpredictable nature of future energy demands, hydrological variability, and technology costs. As note, conventional models often overlook these inherent uncertainties, limiting their ability to inform resilient policy decisions. This limitation is especially problematic for hydropower-dominated systems in Sub-Saharan Africa, where climate change and shifting runoff patterns introduce severe risks to capacity expansion and grid reliability (Carvajal *et al.*, 2019). While some studies employ limited sensitivity analyses, these are insufficient for addressing epistemic uncertainties where historical data cannot reliably predict future outcomes (Koponen and Le Net, 2021). Consequently, there is a pressing need to transition from single-forecast optimizations to Decision-Making under Deep Uncertainty (DMDU) approaches. This study addresses this gap by integrating Robust Decision Making (RDM) with OSeMOSYS to systematically stress-test model outputs across a wide range of plausible futures. By incorporating exploratory modelling

and scenario discovery techniques (PRIM), the research advances knowledge by moving beyond single-scenario optimization toward identifying robust, uncertainty-resilient energy strategies, thereby enhancing the credibility and policy relevance of technoeconomic energy planning.

This study adopts a dual-modelling approach to evaluate energy strategies in Sierra Leone, with a particular focus on power sector development and the expansion of the Bumbuna I Hydroelectric Power Plant.

The research combines three primary tools to validate findings and account for future uncertainties:

OSeMOSYS is an open-source, bottom-up cost-optimization model used to determine the least-cost energy supply mix. It minimizes total discounted system costs while satisfying specified energy demands and technical constraints.

Robust Decision-Making (RDM) is used to stress-test the OSeMOSYS results by running hundreds of variations of key parameters. It employs Latin Hypercube Sampling to evaluate how outcomes change within predefined uncertainty bounds, such as variations in climate-affected water availability, fuel prices, and infrastructure performance.

MAED (Model for Analysis of Energy Demand) is used for demand forecasting and provides the annual energy demand inputs required for the OSeMOSYS analysis.

3. Materials and Methods

3.2 Energy System Model (OSeMOSYS)

3.2.1 OSeMOSYS

The Open-Source Energy Modelling System (OSeMOSYS) is a widely used, open-source energy system optimization framework designed to support long-term energy planning at local, national, and multi-regional scales. Its transparency, flexibility, and relatively low data requirements make it particularly suitable for application in resource-constrained settings and capacity-building contexts (Howells *et al.*, 2011).

OSeMOSYS determines the least-cost configuration of an energy system by optimizing the supply mix—expressed in terms of generation capacity and energy production—required to meet exogenously specified energy service demands over a defined time horizon. The model

minimizes total discounted system costs while satisfying a range of technical, resource, and policy constraints ((Howells *et al.*, 2011).; (GARDUMI, 2016).

The modelling framework represents the energy system as a network of commodities (e.g., biomass, fossil fuels, electricity) and technologies (e.g., generation, transmission, and conversion systems). Energy demands are met through a portfolio of technologies characterized by technoeconomic parameters, including capital costs, fixed and variable operating costs, efficiencies, and operational lifetimes. These technologies utilize resources defined by their availability, technical potential, and associated costs (Howells *et al.*, 2011).

OSeMOSYS can be configured to represent single or multiple energy sectors—including electricity, heat, and transport—across user-defined spatial and temporal resolutions. Policy interventions are incorporated through model constraints, such as emissions limits, renewable energy targets, and capacity expansion restrictions (GARDUMI, 2016).

As with many long-term optimization models, OSeMOSYS assumes a centralized decision-maker, perfect foresight, and fully competitive market conditions. While these assumptions facilitate tractable modelling, they should be considered when interpreting results, particularly in real-world contexts characterized by market imperfections and institutional constraints (Bhattacharyya and Timilsina, 2010).

2.4 Robust Decision, Making Framework

2.4.1 Robust Decision, Making

Robust Decision, Making (RDM) is an iterative, simulation, based decision, support framework designed to evaluate the performance of strategies under conditions of deep uncertainty, where future system states, probabilities, and model structures cannot be reliably characterized (Hall *et al.*, 2012). Rather than identifying a single optimal solution, RDM seeks to identify strategies that perform satisfactorily across a wide range of plausible futures, thereby emphasizing robustness over optimality (Lempert, 2003); (Kwakkel and Haasnoot, 2019).

When integrated with energy system optimization models such as OSeMOSYS, RDM shifts the analytical focus from deterministic, least, cost solutions to robust energy pathways that remain effective under uncertain conditions. This is particularly relevant in energy planning contexts, where key drivers, including fuel prices, hydrological variability, technology costs, and policy environments, are subject to significant uncertainty (Hall *et al.*, 2012).

RDM evaluates the robustness of model outcomes through systematic exploration of uncertainty by varying selected input parameters within predefined bounds and analysing the resulting distribution of outputs. In this study, Latin Hypercube Sampling (LHS) is employed to efficiently generate a large ensemble of scenarios, each representing a plausible future defined by different combinations of uncertain parameters (Mckay, Beckman and Conover, 2000). This exploratory modelling approach enables the identification of vulnerabilities in candidate strategies and provides insights into the sensitivity of system performance to key uncertainties.

Furthermore, RDM supports the identification of robust strategies by highlighting those that exhibit stable performance across a wide range of futures, thereby reducing the risk of policy failure under adverse conditions (Lempert, 2003). In the context of Sierra Leone, uncertain parameters were selected based on their relevance to power sector development and their susceptibility to variability, including hydropower availability, fuel prices, and system performance characteristics.

By integrating RDM with OSeMOSYS, this study enables the systematic stress, testing of modelling assumptions and enhances the robustness, transparency, and policy relevance of energy planning outcomes under deep uncertainty

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2.5 Data Collection and Processing

Data for this study were compiled from a combination of publicly available sources and stakeholder, provided inputs to ensure a comprehensive and context, specific representation of Sierra Leone's energy system. These sources include peer, reviewed literature, feasibility studies, international energy databases, and datasets from previous energy modelling exercises. Additional data were obtained from relevant State, Owned Enterprises (SOEs), including the Electricity Distribution and Supply Authority (EDSA) and the Electricity Generation and Transmission Company (EGTC), with coordination support from the Ministry of Energy. This stakeholder engagement also informed the definition of the model scope and key system boundaries, consistent with best practices in energy system modelling (Howells *et al.*, 2011), 2011; (GARDUMI, 2016).

The dataset was structured to support the implementation of the OSeMOSYS modelling framework, following the Data, to, Deal approach for energy planning and investment analysis (Taliotis *et al.*, 2016). Electricity demand projections were developed using the Model for

Analysis of Energy Demand (MAED), which captures the relationship between socio, economic drivers and energy consumption (*International Atomic Energy Agency (IAEA). (2006).... - Google Scholar, no date*). The MAED model was calibrated using baseline demand estimates derived from Sierra Leone's Integrated Resource Plan (IRP) and other national energy assessments.

Historical electricity generation data were obtained from EDSA and EGTC and subsequently adjusted to account for transmission and distribution losses to estimate final electricity demand. These calibrated demand values were then used as inputs to the MAED model to generate long, term demand projections.

Given data limitations typical of developing country contexts, gaps in country, specific data were addressed by regional proxies and internationally benchmarked datasets. All data were systematically processed, validated, and harmonized to ensure consistency with the input requirements of the OSeMOSYS and MAED models. This integrated data development approach enhances the reliability and transparency of the modelling framework while supporting robust energy system analysis under data, constrained conditions (Bhattacharyya and Timilsina, 2010).

2.6 Scenario Construction and Assumptions

A set of seven scenarios was developed to evaluate alternative pathways for the expansion of the Bumbuna hydropower system and the broader evolution of Sierra Leone's electricity sector under varying infrastructure and technology configurations. Scenario based analysis is widely used in energy system modelling to explore uncertainties and assess the implications of different development trajectories (Bhattacharyya and Timilsina, 2010); (Welsch *et al.*, 2014).

Scenario 1 represents the Business, as, Usual (BAU) case and serves as the baseline against which all other scenarios are compared. In this scenario, the Bumbuna Hydroelectric Power Plant remains at its existing capacity of 50 MW, and no additional expansion is undertaken. The model optimizes the energy system based on current infrastructure, technoeconomic parameters, and projected demand.

Scenarios 2 and 3 explore incremental and large, scale expansion options for Bumbuna I. Scenario 2 reflects planned government interventions, increasing installed capacity by 50 MW to a total of 100 MW through the addition of new turbine capacity. Scenario 3 considers a more

ambitious expansion, increasing capacity by 160 MW to 210 MW while maintaining the existing reservoir constraints, thereby testing the system's ability to accommodate higher generation levels under fixed hydrological conditions.

Scenarios 4 and 7 introduce hybrid system configurations by integrating floating solar photovoltaic (PV) capacity with hydropower infrastructure. Scenario 4 builds on Scenario 2 by adding 25 MW of co, located solar PV, while Scenario 7 extends Scenario 6 by incorporating the same level of solar capacity. Such hybridization strategies have been shown to improve system flexibility, enhance generation during dry seasons, and reduce reliance on hydrological conditions (Cazzaniga *et al.*, 2018).

Scenarios 5 and 6 incorporate the development of the Bumbuna II project at Yiben, introducing additional upstream storage and generation capacity. Scenario 5 combines an 88 MW expansion at Bumbuna I with 55 MW from Bumbuna II, while Scenario 6 adopts a more moderate configuration of 50 MW at Bumbuna I alongside 55 MW from Bumbuna II. These scenarios assess the role of increased reservoir storage in improving seasonal reliability and mitigating hydrological variability, which is a critical factor in hydropower, dominated systems (Carvajal *et al.*, 2019).

All scenarios were implemented within the OSeMOSYS framework, allowing the model to determine the least, cost system configuration subject to the defined infrastructure assumptions and constraints. This structured scenario design enables a systematic comparison of expansion strategies and provides a foundation for subsequent robustness analysis under uncertainty (Howells *et al.*, 2011). Schematic representations of the scenarios are described in Figure 1.

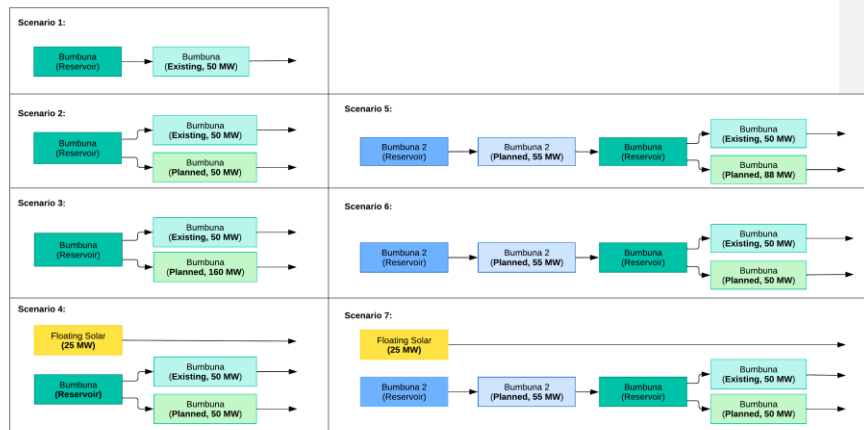


Figure 1. Illustration of the seven key scenarios developed by the Ministry of Energy in collaboration with CCG through stakeholder engagement. Arrows represent the flow of energy across the system, from primary resources through conversion technologies to final electricity demand.

2.1 Reference Energy System

The Reference Energy System (RES) represents a stylized abstraction of Sierra Leone's electricity system, capturing the transformation of primary energy resources into final electricity demand within the OSeMOSYS framework. The RES is constructed in accordance with current system characteristics and planned policy developments. A schematic representation of the model structure is provided in Figure 2, following a standard OSeMOSYS classification.

Primary energy supply consists of domestic renewable resources, biomass, solar, and hydropower, and imported fossil fuels, namely heavy fuel oil (HFO) and light fuel oil (LFO). These resources are converted into electricity through a portfolio of generation technologies defined by technoeconomic parameters. The system boundary also includes cross, border electricity trade, with imports from Côte d'Ivoire via the CLSG interconnection and prospective exports to Guinea.

Electricity produced by these technologies is transmitted through the national grid and represented as a secondary energy carrier. Final electricity demand is disaggregated across key end, use sectors, including residential, commercial, and industrial consumers. The RES

structure enables the model to evaluate least, cost capacity expansion and dispatch decisions subject to resource, technical, and policy constraints.

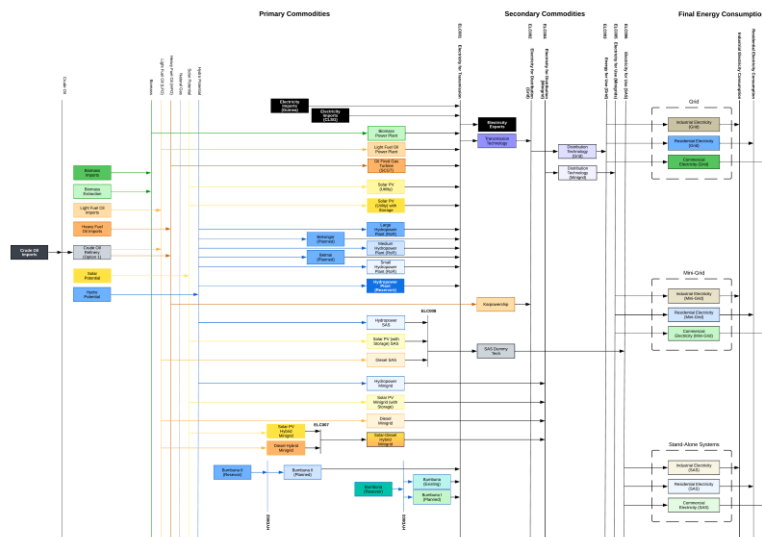


Figure 2: Sierra Leone Reference Energy System

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3 Results

Electricity generation increases across all scenarios over the modelling horizon, driven primarily by sustained growth in electricity demand. As illustrated in Figure 3, total annual generation expands significantly, with the technology mix varying across scenarios depending on infrastructure investments and resource availability. Under several scenarios, the electricity system approaches near 100% renewable generation by 2030, largely due to the increased contribution of hydropower from the Bumbuna system. This transition is particularly pronounced in scenarios incorporating the Yiben reservoir, which enhances storage capacity and enables more consistent generation during dry seasons. Such findings are consistent with previous studies highlighting the critical role of hydropower storage in enabling high renewable energy penetration in developing power systems (Carvajal *et al.*, 2019).

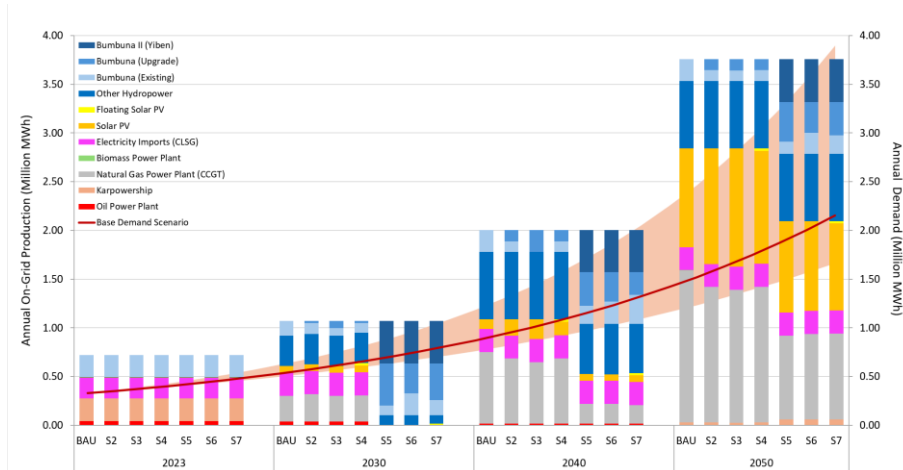


Figure 3: Annual Power Production (Millions MWh)

A substantial reduction in fossil fuel, based generation is observed in Scenarios 5, 6, and 7 (Figure 4). This reduction is primarily attributed to improved hydrological regulation and increased water availability resulting from the Yiben reservoir, which reduces reliance on thermal generation during periods of low inflow. In contrast, the addition of 25 MW of floating solar photovoltaic (PV) capacity has a relatively limited impact on total system generation, suggesting that, within the current system configuration, hydropower remains the dominant driver of supply. This outcome reflects the importance of resource complementarity and system scale when integrating variable renewable technologies (Welsch *et al.*, 2014).

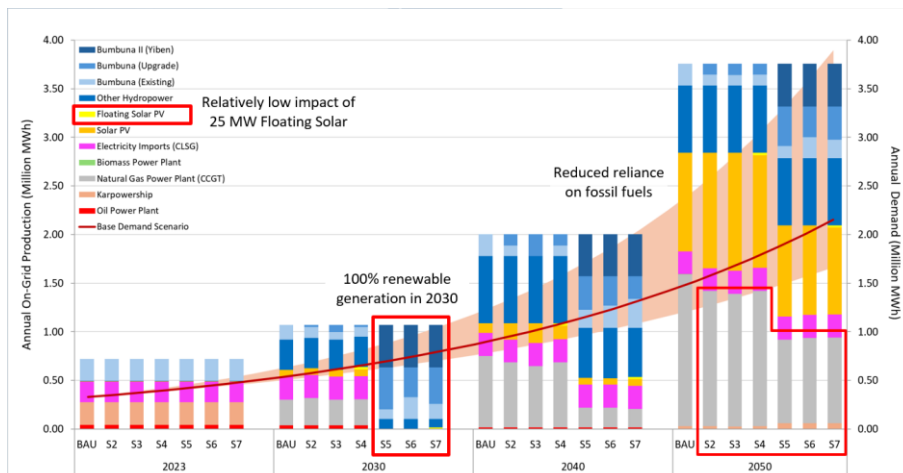


Figure 4: Annual Power Production (Millions of MWh) showing Renewable Generation, Reduction in fossil fuels and the Impact of a 25 MW Floating Solar

Figure 5 further illustrates the evolution of installed capacity across scenarios. The results indicate that the Karpowership is phased out by 2030 in all scenarios, largely due to increased domestic generation capacity from hydropower projects such as Bumbuna expansion, Betmai, Bekongor, and Singimi. However, in Scenarios 2, 3, and 4, natural gas, fired generation is introduced to replace this capacity and meet rising demand, reflecting the need for dispatchable backup in the absence of additional storage.

Notably, scenarios incorporating the Yiben reservoir (Scenarios 5, 6, and 7) require lower total installed capacity compared to those without it. This is due to enhanced system reliability and improved capacity factors of hydropower, which reduce the need for additional generation assets. Across all scenarios, solar capacity expands significantly by 2050, driven by declining technology costs and increasing demand, consistent with global trends in renewable energy deployment (IRENA, 2020).

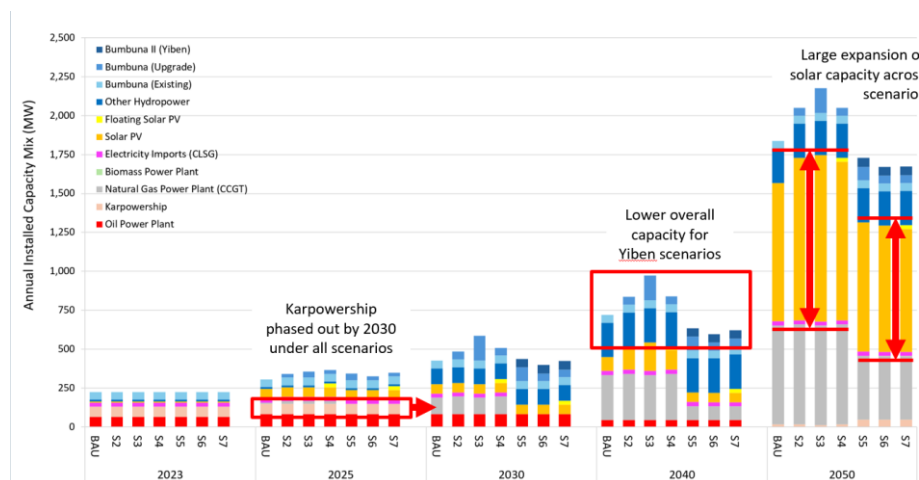


Figure 5. Annual Installed Capacity (GW)

Figure 6 presents a comparative assessment of total system costs across all scenarios over the 2024–2050 period, alongside the corresponding share of renewable electricity generation in 2050. The results indicate that scenarios incorporating the Yiben reservoir, particularly Scenarios 6 and 7, achieve lower overall system costs relative to scenarios focused solely on the expansion of Bumbuna I (Scenarios 2, 3, and 4). This outcome is primarily driven by

reduced dependence on fossil fuel, based generation, leading to significant savings in fuel and operational costs. The increased storage capacity associated with the Yiben reservoir enhances hydropower availability during dry seasons, thereby improving system reliability and reducing the need for expensive thermal backup generation. This finding aligns with existing literature emphasizing the cost, effectiveness of hydropower with storage in minimizing long, term system costs under variable supply conditions (Welsch et al., 2014).

In contrast, Scenario 5 exhibits substantially higher total system costs despite achieving high renewable penetration. This is largely attributable to the significant upfront capital investment required for the combined 143 MW hydropower expansion at Bumbuna and Yiben. While such large, scale investments can deliver long, term benefits, their cost, effectiveness depends on system utilization rates and demand growth trajectories (IRENA, 2020). Overall, the results highlight the importance of balancing capital, intensive infrastructure investments with operational savings, and demonstrate that moderate, well, integrated expansion pathways can yield more cost, efficient outcomes in the long term.

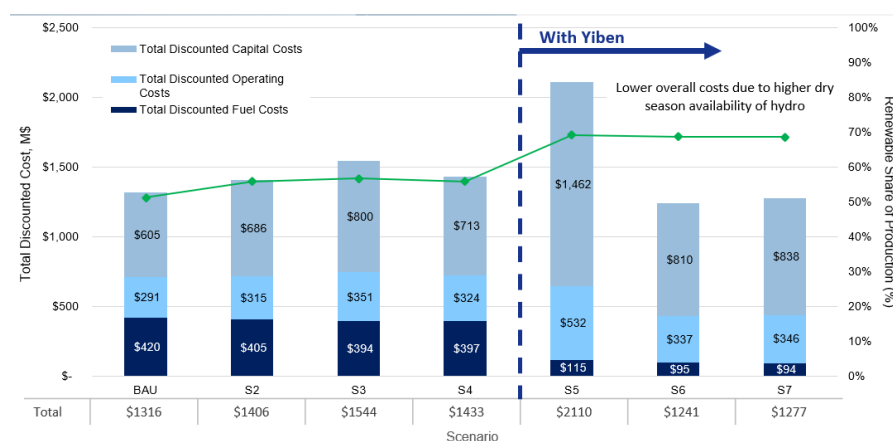


Figure 6: Model Period Costs

Figure 7 presents the outcomes of the Robust Decision, Making (RDM) analysis, illustrating the performance of each scenario across a wide range of plausible futures. The x, axis represents the Levelised Cost of Electricity (LCOE), while the y, axis shows the share of renewable electricity generation in 2050. Each point corresponds to a distinct future generated through systematic variation of uncertain input parameters, with scenarios differentiated by colour. This exploratory approach enables the evaluation of scenario performance under deep uncertainty rather than relying on a single deterministic outcome (Lempert et al., 2006; Kwakkel & Haasnoot, 2019).

The results demonstrate that Scenarios 6 and 7 consistently cluster within the region characterized by low LCOE and high renewable energy penetration. This concentration indicates that these scenarios perform well across a broad range of uncertain conditions, exhibiting both economic efficiency and environmental sustainability. In contrast, other scenarios display a wider dispersion of outcomes, suggesting greater sensitivity to variations in key parameters such as hydropower availability, fuel prices, and demand growth.

The superior performance of Scenarios 6 and 7 can be attributed to the inclusion of the Yiben reservoir, which enhances hydropower reliability and reduces dependence on costly thermal generation. By improving system flexibility and mitigating the impacts of hydrological variability, the reservoir contributes to more stable and cost-effective system outcomes. These findings are consistent with the core principles of RDM, which emphasize the identification of strategies that remain robust, rather than optimal, across a wide ensemble of uncertain futures (Lempert et al., 2006).

Overall, the RDM analysis validates the OSeMOSYS modelling results by demonstrating that scenarios incorporating additional storage capacity not only perform well under baseline assumptions but also maintain their advantages under uncertainty. This strengthens the credibility of the modelling framework and highlights the value of integrating exploratory modelling approaches into energy system planning (Kwakkel & Haasnoot, 2019).

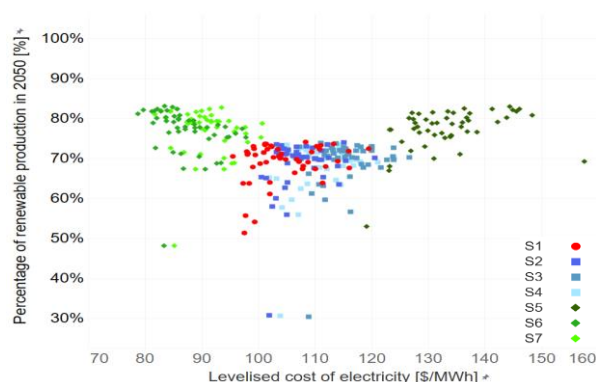


Figure 7: Robust Decision Making

In Figure 8 The model captures only direct (combustion, based) emissions from fossil fuel electricity generation, which explains the observed zero emissions in Scenarios 5, 6, and 7 during the 2026–2035 period. In these scenarios, electricity demand is fully met by renewable sources, primarily hydropower, supplemented by solar and imports, thereby displacing thermal

generation. This reflects a typical outcome in least, cost optimisation models where low marginal cost renewables dominate when sufficient capacity and system reliability are ensured.

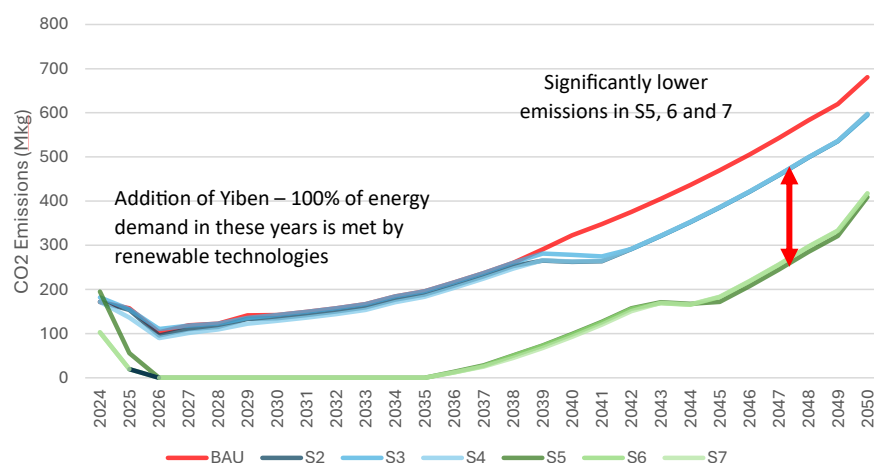


Figure 8: CO2 Emissions from Power Plants

The key driver of emission reductions is the inclusion of the Yiben reservoir, which enhances hydropower availability during dry seasons. This reduces reliance on fossil fuel, based backup generation, which is otherwise required in scenarios without sufficient storage. As a result, Scenarios 5–7 exhibit structurally lower emissions across the modelling horizon.

However, it is important to note that these results represent operational emissions only and exclude lifecycle emissions (e.g., infrastructure construction). Therefore, while the scenarios demonstrate strong decarbonisation potential, they may underestimate total system, wide emissions. Overall, the findings highlight the critical role of storage, enhanced hydropower in enabling low, emission power systems.

4. Discussion and Policy Implications

This study integrates deterministic energy system optimisation using OSeMOSYS with a Robust Decision, Making (RDM) framework to evaluate alternative development pathways for Sierra Leone's power sector under deep uncertainty. The combined approach provides important insights into both the cost, optimal configuration of the system and the resilience of these configurations across a wide range of plausible futures. While OSeMOSYS identifies least, cost solutions under predefined assumptions, the RDM analysis extends this by stress,

testing these solutions, thereby addressing a key limitation of deterministic modelling approaches (Howells et al., 2011; Lempert et al., 2006).

The scenario analysis highlights the critical role of hydropower, particularly when supported by additional storage, in driving a low, cost and low, carbon electricity system. Scenarios incorporating the Yiben reservoir (Scenarios 5–7) demonstrate clear advantages in terms of reduced system costs, lower reliance on fossil fuel, based generation, and improved system reliability. These findings underscore the importance of storage, enhanced hydropower in mitigating seasonal variability and ensuring stable electricity supply in hydrologically dependent systems, consistent with previous studies on renewable, dominated energy transitions (Carvajal et al., 2019).

However, the results also reveal important trade, offs between capital investment and long, term system performance. While large, scale expansion pathways, such as Scenario 5, achieve high renewable penetration, they are associated with significantly higher upfront costs. In contrast, more moderate and integrated expansion strategies (e.g., Scenarios 6 and 7) provide a more balanced outcome, combining cost efficiency with high renewable shares. This suggests that phased and system, oriented investment strategies may be more appropriate for developing countries facing financial and institutional constraints.

The RDM analysis further strengthens these conclusions by demonstrating that Scenarios 6 and 7 are not only cost, effective under baseline assumptions but also robust across a wide range of uncertain futures. Their consistent performance, characterized by low Levelised Cost of Electricity (LCOE) and high renewable energy penetration, indicates reduced sensitivity to key uncertainties such as hydrological variability, fuel prices, and demand growth. This reinforces the value of incorporating flexibility and storage into system design as a hedge against uncertainty (Kwakkel & Haasnoot, 2019).

From a policy perspective, these findings suggest that energy planning in Sierra Leone should move beyond deterministic least, cost optimisation toward uncertainty, aware decision frameworks. The integration of RDM with OSeMOSYS provides a practical pathway for achieving this by enabling policymakers to identify strategies that are both economically efficient and resilient. Investments in hydropower expansion with enhanced storage capacity, such as the Yiben project, should be prioritized, given their demonstrated ability to improve system robustness and reduce long, term costs.

Furthermore, while solar PV deployment increases over time, its relatively limited short, term impact highlights the need for complementary investments in grid flexibility and storage to fully realize its potential. Policymakers should therefore consider integrated planning approaches that combine hydropower, solar, and transmission infrastructure to enhance system resilience and support future demand growth.

Overall, this study contributes to the growing body of literature advocating for the integration of exploratory modelling approaches into energy system planning. By combining OSeMOSYS with RDM, it demonstrates how robust, policy, relevant insights can be generated even in data, constrained environments. This approach is particularly relevant for Sub-Saharan Africa, where uncertainty is high and the consequences of infrastructure investment decisions are significant.

5. Conclusions

This study presents an integrated framework combining OSeMOSYS, based energy system optimisation with Robust Decision, Making (RDM) to evaluate hydropower expansion pathways in Sierra Leone under deep uncertainty. The results demonstrate that while deterministic modelling provides valuable insights into least, cost system configurations, its outcomes can be sensitive to underlying assumptions. By incorporating RDM, this study enhances the robustness and policy relevance of energy planning by systematically evaluating system performance across a wide range of plausible futures.

The analysis shows that hydropower expansion, particularly when coupled with additional storage capacity from the Yiben reservoir, plays a pivotal role in achieving a cost, effective and low, carbon electricity system. Scenarios incorporating Yiben (notably Scenarios 6 and 7) consistently outperform alternatives in terms of lower system costs, reduced reliance on fossil fuel, based generation, and higher renewable energy penetration. Importantly, the RDM results confirm that these scenarios remain robust under varying assumptions, exhibiting stable performance across uncertainty dimensions such as hydrological variability, fuel prices, and demand growth.

In contrast, scenarios focused solely on expanding Bumbuna I without additional storage show greater dependence on thermal generation and higher sensitivity to uncertainty. While large, scale expansion options can achieve high renewable shares, their economic viability is constrained by significant upfront capital requirements. These findings highlight the importance of balanced and flexible investment strategies that optimize both cost and system resilience over the long term.

From a policy perspective, the results underscore the need to move beyond deterministic planning approaches toward uncertainty, aware frameworks that explicitly account for system vulnerabilities. The integration of OSeMOSYS with RDM provides a practical and scalable methodology for achieving this, particularly in data, constrained environments. For Sierra Leone, prioritizing hydropower development with enhanced storage capacity, alongside complementary investments in renewable energy and grid infrastructure, can support a more resilient and sustainable energy transition.

Overall, this study contributes to the advancement of energy planning methodologies by demonstrating how combining technoeconomic optimisation with exploratory modelling can generate robust, decision, relevant insights. The approach is transferable to other developing country contexts facing similar challenges of uncertainty, limited data availability, and the need for long, term infrastructure investment planning.

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